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Development and validation of a risk score for predicting atrial fibrillation recurrence after a first catheter ablation procedure – ATLAS score

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Aims Several predictors of relapse after catheter ablation of atrial fibrillation (AF) have been established, but assessing each patient's individual risk remains challenging. Our aim was to develop and validate a score to estimate the risk of AF recurrence after the first radiofrequency pulmonary vein isolation (PVI) procedure.

Methods and results Independent predictors of AF relapse were identified retrospectively in a two-centre registry of 1934 patients who underwent a first PVI procedure. Using the Cox regression hazard ratios of designated variables, a risk score was developed in a random sample of 50% of the patients (development cohort) and validated in the remaining (validation cohort) half. The accuracy and discriminative power of the predictive model were assessed in both subgroups. During a follow-up of 4.2 ± 2.7 years, 522 patients (27%) relapsed. Five independent predictors of AF recurrence were identified and included in the score: age >60 years (1 point), female sex (4 points), non-paroxysmal AF (2 points), current smoking (7 points) and indexed left atrial volume (1 point for each 10 mL/m^2). The score showed good discriminative power (censored c-statistic of 0.75 in both cohorts). In the development group, AF relapse rates were 8, 11, and 17%/year for low (<6 points), intermediate (6–10 points), and high-risk patients (>10 points), respectively ($P < 0.001$). In the validation group, AF recurrence rates were 8, 11, and 18%/year, respectively ($P < 0.001$).

Conclusion A simple risk score to estimate the rate of AF recurrence after ablation was developed and validated. An external assessment of its usefulness as a patient selection tool seems warranted.

Keywords Atrial fibrillation • Atrial fibrillation recurrence • Pulmonary vein isolation • Radiofrequency catheter ablation • Prognosis • Risk score

Introduction

Pulmonary vein isolation (PVI) is now a well-established treatment for selected patients with atrial fibrillation (AF).[1] However, despite high early success rates, AF recurrence is common during follow-up, offsetting the potential benefits of the procedure.[2–4] Identifying the patients who are unlikely to remain in sinus rhythm seems a logical way to improve success rates and avoid the unnecessary risks and costs of ineffective ablations. While several predictors of AF relapse have been identified,[2,3] assessing each patient's individual risk of recurrence remains challenging, in part due to the absence of a risk stratification tool that would condense the prognostic information conveyed by several different variables. The purpose of this study was to develop and validate a score to assess the risk of AF recurrence after a first radiofrequency PVI procedure.

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What's new?

- Development and validation of a clinical risk score to estimate the risk of atrial fibrillation (AF) recurrence after a first catheter ablation procedure.
- Identification of long-term independent predictors of AF recurrence in a large dataset of patients.
- Recognition of left atrial volume as the main determinant of AF recurrence after catheter ablation.

Methods

Patient population and study design

All consecutive patients with symptomatic drug-refractory AF undergoing percutaneous PVI in two Portuguese centres (Hospital Santa Cruz, Carnaxide, Portugal; and Hospital da Luz, Lisbon, Portugal) were included in an observational registry, which we used for this retrospective study. The first centre enrolled patients between May 2005 and November 2015, and the second centre between June 2007 and September 2015. From a pool of 2836 procedures, we excluded patients with previous AF ablations ($n = 428$), and those without 3D quantification of LA volume neither by computed tomography (CT) nor electroanatomical mapping ($n = 352$). The latter was used as an exclusion criterion, since left atrial (LA) volume has been shown to be a powerful prognostic marker in this setting.[5,6] Patients who were lost to follow-up less than 6 months after the procedure were also excluded ($n = 122$). The final study population ($n = 1934$) was randomized on a 1:1 ratio into development and validation cohorts. In the development cohort, Cox regression analysis was used to identify independent predictors of clinical and/or electrocardiographic AF recurrence. The β regression coefficients of the selected variables were then used to develop a risk score whose accuracy and discriminative power were evaluated in the validation cohort.

Atrial fibrillation was categorized as paroxysmal if self-terminated in <7 days, persistent if episodes lasted ≥7 days or required cardioversion, or long-standing persistent if AF was maintained for more than 12 months.[1] The majority of patients ($n = 1332$, 69%) underwent a 64-slice cardiac CT scan less than 48 h before the ablation procedure for the assessment of pulmonary vein anatomy, measurement of LA volume, exclusion of thrombi, and integration with electroanatomical mapping.[6] Left atrial volume was calculated by tracing the LA borders on CT images, excluding the pulmonary veins and the left atrial appendage. In patients where cardiac CT could not be performed (mainly for logistical reasons), LA volume was estimated from CARTO® (Biosense Webster Inc., Diamond Bar, CA, USA) electroanatomical mapping at the time of ablation. Left atrial volumes were indexed to body surface area (BSA). In patients who had LA volume estimated by different methods (2D echocardiogram, cardiac CT and/or electroanatomical mapping), a correlation analysis was performed to assess comparison between all methods.

Pulmonary vein isolation protocol

Pulmonary vein isolation was guided by electroanatomical mapping, using either NavX® (St Jude Medical® Inc, St Paul, MN, USA) or CARTO® systems. The right femoral vein was used as the preferred vascular access, through which three catheter electrodes were introduced: (i) a decapolar catheter, advanced through the coronary sinus; (ii) a variable circular mapping catheter, placed in the pulmonary veins; and (iii) an irrigated-tip ablation catheter. Left atrial access was established by trans-septal puncture. Radiofrequency ablation was performed more than 5 mm from the PV ostia, with continuous lesions enclosing the left and right pairs of PV. At Hospital Santa Cruz, all patients underwent conventional manually guided ablation, while at Hospital da Luz a Niobe II magnetic navigation system (Stereotaxis® Inc., St. Louis, MO, USA) was used. If typical atrial flutter was previously documented or observed at the time of the procedure, an additional cavotricuspid isthmus (CTI) ablation was performed ($n = 164$ patients). The treatment was considered successful if complete electrophysiological PVI was achieved. When required, an electrical cardioversion was performed at the end of the procedure. Oral anticoagulation was resumed 6 h after the ablation, maintained for 6 months and then withdrawn or continued according to CHA_2DS_2-VASc criteria ($CHADS_2$ before 2009). Patients on vitamin K antagonists with subtherapeutic international normalized ratio (INR) were kept on subcutaneous enoxaparin 1 mg/kg every 12 h until an adequate INR was achieved. As a general rule, class I/III antiarrhythmic drugs were maintained in all patients for the first 3 months after the procedure and then withdrawn if there was no AF recurrence. A proton pump inhibitor was also prescribed for the first month after the ablation.

Study endpoint and patient follow-up

The study endpoint was AF recurrence, defined as symptomatic or documented AF and/or other atrial arrhythmias, after a 3-month blanking period. Symptomatic AF was defined as the presence of symptoms considered to be likely due to AF episodes. Documented AF was defined by the presence of at least one episode of AF lasting more than 30 s in any electrocardiogram (ECG), 24 h Holter monitoring or event-loop recording. The follow-up protocol comprised outpatient visits with 12-lead ECG and 24 h Holter monitoring on the 1st, 3rd, 6th, and 12th months post-ablation, followed by yearly assessments. Patients were encouraged to contact the department if they experienced symptoms of AF recurrence. Whenever clinical records were insufficient, a structured telephonic interview was conducted. Patients who were kept on antiarrhythmic drugs after the 3rd month of follow-up were not considered as failed ablation.

Statistical analysis

Normally and non-normally distributed variables were expressed as mean ± standard deviation and median, respectively. Differences between groups were assessed using independent samples *t*-test and Fisher's exact test for continuous and categorical variables, respectively. The study population was split into development and validation cohorts by generating a random number for each patient and assigning odd and even numbers to each of these groups.

Univariate proportional-hazards Cox regression was used to identify predictors of time to AF recurrence in the development cohort. The following variables were assessed: age, sex, type of AF (paroxysmal vs. non-paroxysmal), indexed LA volume, body mass index, hypertension, diabetes, cigarette smoking, hypercholesterolaemia, known obstructive coronary artery disease, and left ventricular systolic dysfunction (ejection fraction <50%). Smoking was defined as any cigarette consumption during the previous 6 months. In order to simplify clinical score calculations, age was dichotomized using a threshold (60 years) identified with receiver-operating characteristics (ROC) curve analysis.

Pearson's correlation with a two-tailed test of significance was used to evaluate LA volume estimation by different methods.

Variables with a *P*-value ≤0.10 in univariate analysis were entered simultaneously in a multivariate Cox regression model and considered to be statistically significant if $P < 0.05$. Multicollinearity was excluded by assessing Pearson's correlation coefficient between pairs of continuous variables (all <0.60). In order to create a practical risk score, each of the independent predictors was assigned a number of weighted points proportional to its β regression coefficient value (rounded to the nearest

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Table 1 Baseline patient characteristics

	All patients (n=1934)	Development group (n=960)	Validation group (n=974)	P-value
Age (years)	59±11	58±12	59±11	0.04
Male sex—no. (%)	1330 (69)	664 (69)	666 (68)	0.40
Body mass index—kg/m^2 (IQR)	27 (5)	26 (5)	27 (5)	0.53
Type of AF				0.41
Paroxysmal—no. (%)	1488 (77)	734 (76)	754 (77)	–
Persistent—no. (%)	284 (15)	147 (15)	137 (14)	–
Long-standing—no. (%)	162 (8)	74 (8)	88 (9)	–
Indexed LA volume mL/m^2 (IQR)	54 (44-65)	55 (44-65)	54 (44-66)	0.49
Hypertension—no. (%)	770 (40)	380 (40)	390 (40)	0.89
Diabetes mellitus—no. (%)	136 (7)	62 (7)	74 (8)	0.38
Hypercholesterolemia—no. (%)	408 (21)	204 (21)	204 (21)	0.78
Current smoking—no. (%)	188 (10)	94 (10)	94 (10)	0.88
Known coronary heart disease—no. (%)	82 (4)	44 (5)	38 (4)	0.43
LV systolic dysfunction—no. (%)	153 (8)	81 (8)	72 (7)	0.31
Previous stroke—no. (%)	46 (2)	21 (2)	25 (3)	0.66
CHA_2DS_2-VASc—no. (IQR)	2 (2)	2 (2)	2 (2)	0.43
Patients with at least 1 antiarrhythmic drug—no. (%)	1840 (95)	918 (96)	922 (95)	0.46

AF, atrial fibrillation; LA, left atrium; IQR, interquartile range; LV, left ventricle.

integer)—the smallest beta-coefficient was identified and regarded as the least common denominator, by which all others where then divided, yielding the multiplying coefficient for each variable in the score. The resulting score, defined as the sum of the points assigned to the risk factors of each individual, was calculated in both the development and validation cohorts. Score thresholds for defining low, intermediate, and high-risk of relapse were chosen in the development cohort through analysis of (i) log-rank and (ii) χ^2 tests, (iii) statistically significant differences between ATLAS score values, (iv) visual inspection of Kaplan–Meier curves of AF-free survival, (v) pairwise comparison between all risk strata, and (vi) patients representation in each stratum. These thresholds were then applied to the validation cohort.

Differences in actuarial curves were assessed with the log-rank test. The score's discriminative power (i.e. the extent to which the model can differentiate those who will or will not relapse) was evaluated by the censored C-index.[7] Model calibration (reflecting how close predicted recurrence rates are to the actual observed rates) was assessed by the calibration-in-the-large statistic and calibration slope.[8] Statistical analysis was performed using Statistical Package for Social Sciences (SPSS) version 22.0 (SPSS Inc., Chicago, IL, USA) for Mac OS. Statistical significance was set at P <0.05 (two-sided).

Ethical standards

All human and animal studies have been approved by the appropriate ethics committee and have therefore been performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments. The patients signed an informed consent both for the procedure and publication of any relevant data.

Results

The baseline clinical characteristics of the 1934 patients included in this study are summarized in *Table 1*. Overall, the development and validation cohorts were well balanced, except for a small but statistically significant difference in mean age. Regarding LA size assessment, 2D echocardiogram correlated poorly with cardiac CT (n = 227, r = 0.43, P < 0.001) and electroanatomical mapping (n = 416, r = 0.47, P < 0.001). A good correlation between CT and CARTO® volume estimation methods was observed (n = 748, r = 0.77, P < 0.001) (Supplementary material online, *Figure S1*).

During a mean follow-up of 4.2 ± 2.7 years, 522 patients (27%) experienced AF recurrence (259 and 263 in the development and validation groups, respectively, P = 0.88). Less than 5% of these recurrences in both groups were due to atrial arrhythmias other than AF (P = 0.58). The majority of AF relapses were established by both symptoms and electrocardiographic documentation (n = 282; 54%), whereas clinical relapse (n = 177; 34%) and documented silent AF (n = 63; 12%) accounted for the remainder adjudicated recurrences.

Although atrial flutter was previously documented or induced in 164 patients, an additional CTI ablation was performed in 480 patients—criteria based on operator's decision. CTI ablation had no impact in AF-free survival (HR 1.09, 95% CI 0.84–1.41, P = 0.54) and AF annual relapse rates.[9] Maintaining antiarrhythmic drug therapy after the 3-month blanking period had no impact in AF relapse in the subgroup of patients who received CTI ablation.

At the end of the procedure, 433 patients underwent electrical cardioversion.

After the 3rd month of follow-up, 217 patients (11%) were kept on antiarrhythmic drug therapy (109 in the derivation group and 108 in the validation group, P = 0.71), with 52% relapsing in each group (P = 0.76). The vast majority was prescribed class III antiarrhythmic agents—amiodarone [153 (71%) patients], sotalol [33 (15%) patients]—but also a selection of class I C drugs—flecainide [16 (7%) patients], and propafenone [15 (7%) patients]—again without significant differences between both development

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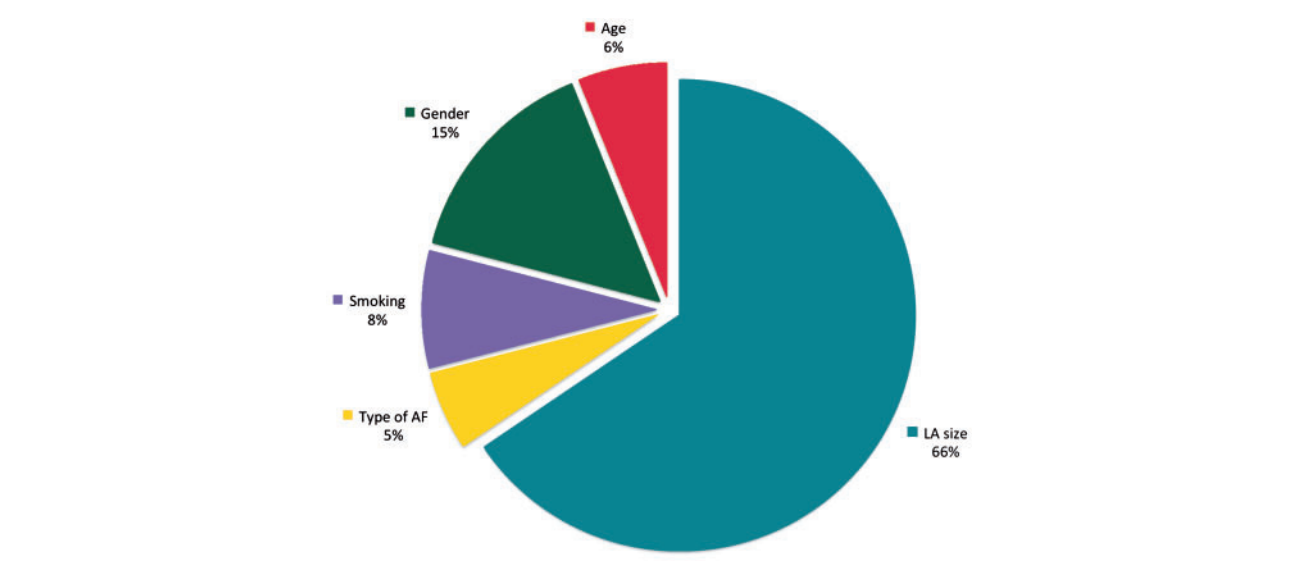

Figure 1 Relative contribution of the five components of the ATLAS score to each individual's final score value.

Table 2 Predictors of time to AF recurrence in univariate and multivariate Cox regression analysis

	Univariate analysis			Multivariate analysis			
	HR *(NA)*	95% CI	*P*-value	(B)	aHR	95% CI	*P*-value
Age >60 years	1.32	1.11–1.57	0.002	0.12	1.25	1.04-1.49	0.017
Female sex	1.19	0.99–1.43	0.06	0.39	1.25	1.03–1.52	0.023
Smoking	1.75	1.31–2.33	<0.001	0.65	1.88	1.40–2.51	<0.001
Non-paroxysmal AF	1.28	1.06–1.55	0.01	0.22	1.24	1.02–1.50	0.032
LA volume (each 10 mL/m^2)	1.01	1.01–1.02	<0.001	0.11	1.12	1.04–1.17	<0.001
Hypercholesterolaemia	1.36	1.1–1.68	0.005	0.19	1.21	0.96–1.51	0.110
Diabetes mellitus	1.46	1.06-2.01	0.02	0.29	1.33	0.96-1.84	0.090
Hypertension	1.05	0.81-1.36	0.696			Not included	
BMI (kg/m^2)	0.75	0.51-1.08	0.12			Not included	
LV systolic dysfunction	0.95	0.45-2.01	0.78			Not included	
Known CAD	1.03	0.55–1.93	0.94			Not included	
CHA_2DS_2–VASc	1.04	0.97–1.11	0.30			Not included	

HR, non-adjusted hazard ratio; aHR, adjusted hazard ratio; CI, confidence interval; (B), β coefficient; AF, atrial fibrillation; LA, left atria; LVSD, left ventricular systolic dysfunction; BMI, body mass index; LV, left ventricle; CAD, coronary artery disease.

Table 3 Calculation of the ATLAS score

Variable	Number of points
Age > 60 years	1
Type of AF (non-paroxysmal)	2
Left **a**trial volume indexed to BSA	1 point per each 10 mL/m^2 (rounded to nearest integer)
Sex (female)	4
Smoking (current cigarette smoking)	7

Example: A 64 years-old non-smoker woman with paroxysmal AF and an LA volume of 57 mL/m^2 would have an ATLAS score of 11 (=1 + 0 + 6 + 4 + 0).

and validation groups. A secondary analysis excluding these patients yielded the same independent predictors of recurrence and similar regression coefficients (Supplementary material online, *Table S1* and *Figure* S2).

The results of Cox regression analysis are presented in *Table 2*. Seven variables associated with AF recurrence were identified in univariate analysis and entered into the multivariate model, yielding five independent predictors of relapse. Each of these variables was assigned a number of points proportional to its β coefficient (*Table 3*). In order to aid memorization, the acronym ATLAS (A, Age; T, Type of AF; LA, LA volume; S, Sex and Smoking) was created. The relative contribution of the five variables to each individual's ATLAS score is

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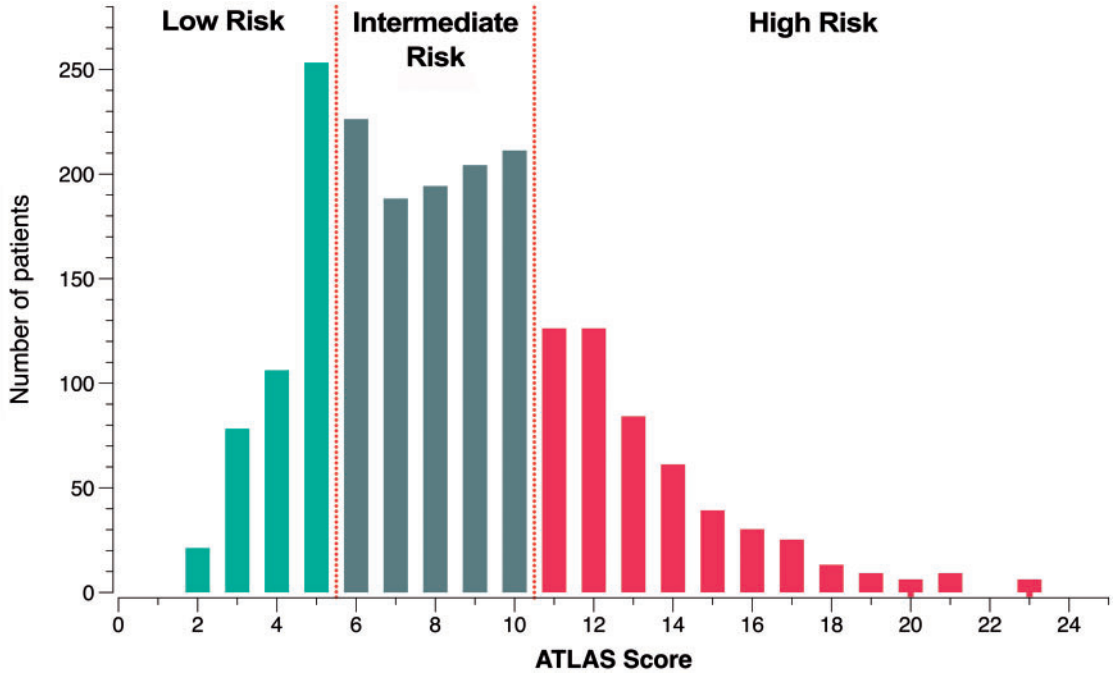

Figure 2 Histogram of ATLAS score values. The vertical dashed lines represent the boundaries between low (<6 points), intermediate (6–10 points), and high-risk categories (>10 points).

presented in *Figure 1*, showing that, on average, indexed LA volume accounts for two thirds of the total value.

The distribution of the ATLAS score in the overall population is depicted in *Figure 2*. The median score was 8 (IQR 6–11), with no significant difference between the development and validation groups (both with median scores of 8, $P = 0.890$).

The score displayed good discriminatory power for earlier AF recurrence, with a censored C-index of 0.75. Three risk categories were defined: low-risk (<6 points), intermediate-risk (6–10 points), and high-risk (>10 points). These categories corresponded to 23.5, 52.5, and 24.0% of the total population, respectively. The AF-free survival curves of low-, intermediate-, and high-risk patients are presented in *Figure 3*. There was a statistically significant difference in AF-free survival between the three risk score categories (log-rank $P < 0.001$ in both the development and validation cohorts). The hazard ratio for intermediate-risk was 1.10 [95% CI 0.90–1.35 (*P*-value 0.35)] and for high-risk 1.60 [95% CI 1.28–2.00 (*P*-value <0.001)].

Within each risk category, there were no statistically significant differences in yearly relapse rates between the development and validation cohorts: 8, 11, and 17%/year for low-, intermediate- and high-risk patients in the development cohort, vs. 8, 11, and 18%/year in the validation cohort.

Analysis of yearly relapse rates showed a significant gradient of risk of AF recurrence across the spectrum of ATLAS score values (*Figure 4*). The calibration-in-the-large statistic (0.077, $P = 0.272$) and a calibration slope of 0.93 showed an overall good calibration, supporting the validity of the prediction model.

The ATLAS score was developed in a population mostly composed of paroxysmal AF patients (77%). Since there is an underrepresentation of non-paroxysmal AF, a subgroup analysis was performed to assess applicability to these patients. The ATLAS score retained its predictive power and ability to identify high-risk patients for AF relapse after a first PVI, regardless of the type of AF (paroxysmal vs. non-paroxysmal) (Supplementary material online, *Figure S3*). The annual relapse rates were similar in the low/intermediate-risk categories (10%/year in both groups) but higher in the high-risk group of patients with non-paroxysmal AF (19%/year vs. 15%/year).

Discussion

Prognosis after pulmonary vein isolation is remarkably heterogeneous, with some patients experiencing early relapse while others remain free of AF for years.[2,10] In this study, we developed and validated a score that estimates the risk of recurrence after a first PVI procedure. Two important aspects of a risk prediction model were assessed: discrimination (the ability to identify those who will relapse) and calibration (reflecting how close predicted recurrence rates are to the actual observed rates). The ATLAS score showed good discriminative power, substantiated in a censored C-statistic of 0.75 and a more than three-fold difference in the yearly rate of AF relapse across the spectrum of score values. Model calibration was similarly good, with a calibration slope of 0.93 supporting the accuracy of the score.

In our cohort, patients who were kept on antiarrhythmic drugs after the 3-month blanking period were not considered failed ablation. Although this remains controversial, the current body of evidence supports that antiarrhythmic drugs are not superior to other agents (e.g. beta-blocker) in maintaining sinus rhythm after PVI.[11,12] Nevertheless,

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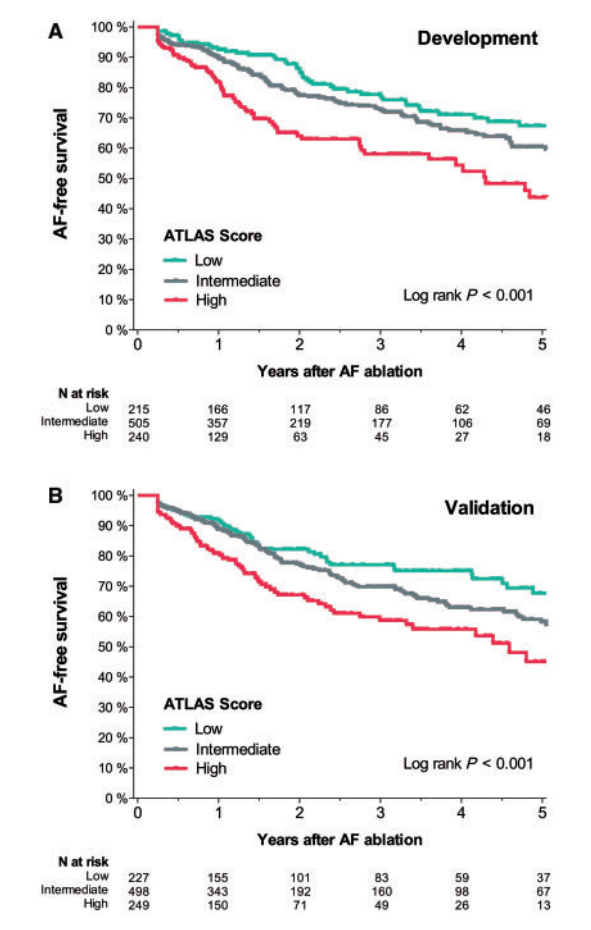

Figure 3 Kaplan–Meier curves of AF-free survival of patients with low, intermediate, and high-risk ATLAS scores in the development cohort (*A*) and validation cohort (*B*).

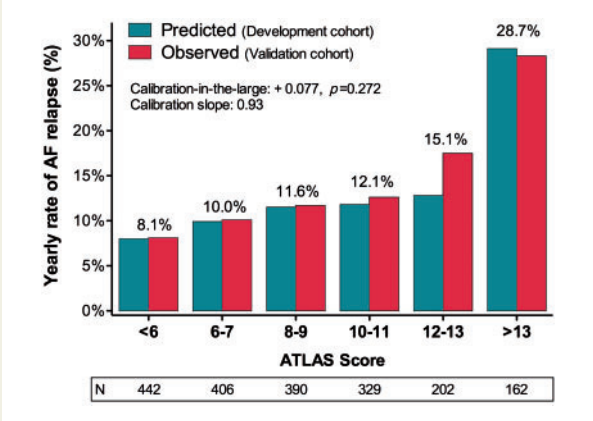

Figure 4 Predicted vs. observed rates of AF relapse across categories of ATLAS scores. The numbers on top of the columns refer to the recurrence rates of the development and validation cohorts combined.

our sensitivity and subgroup analysis demonstrated that this had no impact in the ATLAS score predictive/discriminative power.

Studies assessing AF recurrence after PVI have shown some discordance regarding the predictors of relapse.[2–4] Our analysis yielded five independent predictors of recurrence: LA volume, female sex, cigarette smoking, age, and type of AF (paroxysmal vs. non-paroxysmal). Left atrial volume showed the strongest association with AF-free survival, outweighing all other factors. Although LA size has long been recognized as a prognostic marker, only 4 of the 20 studies included in a recent review found an independent association with relapse rates.[3] However, in those studies LA size was assessed using echocardiographic diameters, which tend to underestimate LA enlargement and correlate poorly with more accurate techniques, such as cardiac CT and magnetic resonance.[13–16] Additionally, a higher inter-observer variability has also been reported with 2D echocardiographic assessment of LA size.[17] It is well recognized that clinical scores demand a trade-off between simplicity and accuracy, but our findings suggest that any prediction tool that does not include LA volume (or an adequate surrogate) assessed by a more precise technique will compromise accuracy in a significant manner. Therefore, we believe that the increase in score complexity associated with adding a reproducible LA volume CT-scan estimation is justified. In studies where LA volume was used, a robust independent association between LA dilatation and AF relapse was found.[5,6] Larger LA volumes probably reflect more extensive remodelling and also larger surface area and greater wall stress and thickness, possibly rendering the tissue more resistant to the creation of transmural lesions. Left atrial dilatation may also increase the likelihood that, once PV reconnection occurs, AF is induced and sustained.

With some surprise, cigarette smoking emerged as an independent predictor of AF recurrence. Although smoking has been previously associated with an increased risk of AF in the general population,[18,19] thus far only one small study showed an association between smoking and AF recurrence after PVI.[20] This relationship is not necessarily causal, but nicotine has been shown to increase atrial vulnerability to fibrillation[21] and induce atrial fibrosis,[22] itself a strong predictor of AF recurrence after ablation.[23] Further studies are warranted to confirm and assess the possible mechanisms of this link.

Gender was also independently associated with AF-free survival in our cohort. Other studies have shown higher recurrence rates in women,[5,24,25] although this was not a universal finding. Later referral, higher incidence of non-PV triggers, and gender-related anatomical differences are some of the possible explanations.[26–28]

To the best of our knowledge, two other scores have been linked to AF recurrence after PVI. The DR-FLASH score was developed to identify patients with low voltage areas on LA electrograms that could require extensive substrate modification instead of PVI alone.[24] Even though it was not the purpose of the score, its association with long-term recurrence of AF was tested in a retrospective cohort of 484 patients, where those with scores >3 were almost twice as likely to experience an AF recurrence within 1 year after ablation. The recently published CAAP-AF score was developed specifically for predicting long-term freedom from AF after ablation. It includes age, type of AF, LA diameter, sex, number of failed antiarrhythmic drugs, and coronary artery disease.[29] Unfortunately, we could not perform a direct comparison between ATLAS and these two scores, since our registry database lacks some of the variables required for its calculation.

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Even though the ATLAS score was devised to estimate the rate of AF relapse and aid decision-making, it is no substitute for sound clinical judgment. There is currently no consensus on what constitutes an acceptable rate of recurrence, and this will probably depend on the clinical setting, procedural risk, and patient preference, among other factors. Consequently, thresholds for classifying patients as low, intermediate or high-risk are largely arbitrary. Users of the ATLAS score might choose to employ other clinically meaningful risk thresholds.

Limitations

Various limitations of this study should be recognized. Atrial fibrillation relapse is most likely underestimated in this population, since recurrences are frequently asymptomatic and our follow-up protocol did not include continuous ECG monitoring. However, since the main goal of AF ablation is to improve symptoms, we believe our pragmatic follow-up protocol captures the essential benefit of the procedure.

Several variables with established prognostic value (e.g. AF burden, left atrial fibrosis, obstructive sleep apnea, etc.) were not systematically evaluated, preventing their inclusion in the model.

Smoking was assessed as a dichotomous variable (and not in pack-year units). Notwithstanding the possible loss of information, this approach replicates the assessment of smoking status in commonly used cardiovascular risk calculators such as the heart SCORE[30] and the ASCVD algorithm.[31]

Different methods were used to measure LA volume, and several patients were lost to follow-up. Assessment of LA volume by cardiac CT may also be regarded as a limitation, since this information is usually unavailable when a patient is being considered for AF ablation. However, in our cohort, this was the single most important prognostic factor.

Although the subgroup analysis showed no significant differences in AF relapse regarding the type of AF, an underrepresentation of non-paroxysmal AF patients in our cohort warrants additional validation in external populations.

Further studies will be required to ascertain whether it can be replaced by a more easily obtainable measure of LA dilatation without significant loss of prognostic information. Finally, although the ATLAS score showed good discriminative power and calibration, its portability to other populations remains to be established.

Despite these limitations, our findings could have important implications for clinical practice. After external validation, the ATLAS score may become a useful tool for clinicians to refine patient selection and inform patient–physician discussions, for researchers to design clinical trials, and for policymakers to improve the allocation of healthcare resources.

Conclusion

We developed and internally validated a risk model to estimate the rate of AF recurrence after a first PVI procedure. The resulting 5-variable score (ATLAS: Age > 60 years, Type of AF, indexed Left Atrial volume, Sex and Smoking) showed good discriminative power and good calibration. If validated in independent datasets, this score might prove a useful tool for selecting patients who will benefit the most from AF ablation, avoiding the unnecessary risks and costs of ineffective procedures.

Supplementary material

Supplementary material is available at *Europace* online.

Conflict of interest: none declared.

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